AI in signal and audio processing

Thomas de Jaeger

thomas.de-jaeger@epita.fr

Spring 2025

Class details

- 24h: 3h*8
- Grade: 50% Labs
  50% Final Project
- All the material will be online: https://github.com/tdejaeger/Epita_lectures
- For anonymous feedback: Feedback - Google Docs

Class organisation

- Lectures (30-1h each class)
- Labs during class
- Project

Final Project Overview — COVID19 detection

- **Project Goal:** Build a complete AI pipeline that predict **COVID19** using audio processing and deep learning techniques.

- You are free to analyze the data as you want. You can work alone or in group (max 2p).
- Output: Presentation (10-15min)+ jupyter notebook/google colab

Data: Coswara: A respiratory sounds and symptoms dataset for remote screening of SARS-CoV-2 infection | Scientific Data

Grade:
- /5 Work, results, analysis
- /5 Presentation, oral expression, slides
- /5 Critical thinking
- /3 How you answer the questions
- /2 the questions you will ask during other presentations

Syllabus

- **Session 1:** Introduction & Fundamentals of Signal/Audio
- **Session 2:** Audio Features & Librosa
- **Session 3:** Audio Processing with Torchaudio
- **Session 4:** Sound Detection and Classification
- **Session 5:** Sequence Modeling with RNNs and LSTMs
- **Session 6:** Automatic Speech Recognition (ASR)
- **Session 7:** Speaker and Emotion Recognition
- **Session 8:** Project presentation

Environment & Requirements

Tools & Libraries to Install

Session	Topic	Tools / Libraries
1	Fundamentals of Signal & Audio Processing	numpy, scipy, matplotlib
2	Audio Features & Librosa	librosa, scikit-learn, matplotlib
3	Audio Processing with Torchaudio	torchaudio, PyTorch
4	Sound Classification	librosa, scikit-learn, PyTorch, matplotlib
5	Sequence Modeling (RNN, LSTM)	PyTorch, torchaudio
6	Automatic Speech Recognition (ASR)	torchaudio, transformers, Wav2Vec2, Whisper
7	Speaker & Emotion Recognition	librosa, PyTorch, scikit-learn

You can run this course on **Google Colab** or **locally**.

Google Colab

What is Google Colab? Google Colab is a free cloud service that supports Jupyter notebooks and provides free access to computing resources including GPUs.

To use it:

- Log in with a Google account
- Open a .ipynb notebook from Drive or GitHub, or create a new one
- To enable GPU: Runtime > Change runtime type > GPU

Why we use it:

- Free and powerful
- No local install needed
- Familiar interface for Jupyter users

Upload files as a drive from your local machine

```
from google.colab import files
input_file = files.upload()
```